

SING-HAO KU

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RESEARCH INTERESTS

Health Economics · Population Studies · Applied Econometrics · Labor and Social Policy

EDUCATION

New York University (NYU)

M.A. in Economics; GPA: 3.81/4.00

New York, NY

09/2024 – 05/2026

- Concentration: Econometrics and Quantitative Economics
- Relevant Coursework: Applied Statistics and Econometrics, Machine Learning for Economics, Rationality and Behavioral Economics, Empirical Research Methods, Messy Data and Machine Learning

National Taiwan University (NTU)

B.B.A. in International Business, Minor in Economics; GPA: 3.75/4.30 (3.65/4.00)

Taipei, Taiwan

09/2019 – 06/2023

MASTER'S THESIS

Coverage Substitution After Job Loss: How the ACA Altered Post-Displacement Insurance Trajectories

M.A. Thesis · Advisor: Prof. Gustavo Pereira · New York University · Highest Distinction in Cohort

01/2026 – 05/2026

[\[Full paper\]](#)

- Designed and executed a propensity-score-matched event-study difference-in-differences on a person-month panel of 5.9 million observations (MEPS 2004–2018), with a two-part hurdle decomposition of premium burden and latent class analysis of pre-displacement job profiles.
- Found the ACA absorbed roughly one-quarter of the two-month coverage loss and one-third of the four-month loss, with gains concentrated in a high-coverage Finance and Professional class while three precarious-work classes showed essentially no gains.

RESEARCH EXPERIENCE

NYU Silver School of Social Work

Research Assistant to Prof. Wen-Jui Han

New York, NY

06/2025 – 05/2026

Research focuses on how parental employment affects adolescent cognitive development and risk-taking behaviors.

- Developed a reproducible data-cleaning workflow in Stata for the National Longitudinal Survey of Youth 1979 (NLSY79) and its Children Supplement, cleaning approximately 11,000 records across 15 waves and constructing over 50 derived variables; implemented a multiple imputation workflow to handle missing data across waves.
- Implemented sequence cluster analysis to identify maternal employment trajectory patterns, then applied growth-curve modeling and stage-specific regression analysis to examine how nonstandard and precarious parental work schedules influence adolescent cognitive development and risk-taking behaviors.

CUNY Graduate School of Public Health

Research Assistant to Prof. Mustafa Hussein

New York, NY

06/2025 – 05/2026

Research focuses on the impact of healthcare policy changes on insurance coverage and out-of-pocket premium payments.

- Built an end-to-end Stata pipeline integrating 21 years of MEPS Full-Year Consolidated, Jobs, and Person-Round files into a person-month panel of 301,000 individuals, constructing measures of insurance status, employment, wages, and medical expenditures.
- Conducted imputation and data wrangling across the panel to support causal-inference analyses of healthcare policy and labor outcomes.

Quanton Corporation

Research Assistant (Full-time)

Taipei, Taiwan

07/2023 – 07/2024

Think tank and data-science research consultancy providing policy analysis and business intelligence.

- Analyzed government and consumer survey data in Stata to evaluate policy initiatives and produce executive-level visualizations and reports for government officials and commercial clients.
- Designed survey instruments in Qualtrics, and assisted with a methodology paper on Random Forest imputation for structural changes in longitudinal survey design, later presented at an AAPOR-affiliated venue.

SELECTED ACADEMIC PROJECTS

Determinants of Household Debt Across OECD Economies

Group Project (Technical Lead) · “Messy Data and Machine Learning” final project

09/2025 – 12/2025

- Applied Random Forest, LASSO, and multinomial logistic regression; Random Forest outperformed multinomial logistic regression (90.5% vs. 76.9% accuracy), identifying credit market accessibility and income inequality as dominant predictors of household debt levels.

LLM-Based Author Disambiguation on Eugenics-Movement Literature

“Empirical Research Methods” project

09/2025 – 12/2025

- Cleaned and standardized large-scale bibliometric records on the eugenics movement (1880–1945) and refined a domain-specific lexicon to structure inconsistent historical source data.
- Built an R pipeline calling the Gemini API to disambiguate and correct author and source records at scale, designing blocking and pairwise-comparison routines for unstructured data.

Medicaid Expansion and Out-of-Pocket Costs: A Double Machine Learning Approach

Author · “Machine Learning for Economics” term paper

New York, NY

02/2025 – 05/2025

- Applied PLIV estimator in R using DML (Random Forest, LASSO, XGBoost with cross-fitting) to estimate the causal effect of Medicaid expansion on medical out-of-pocket spending.

Assessing the Financial and Coverage Effects of Medicaid Expansion

Group Project Lead (Technical) · “Applied Statistics and Econometrics” final project

New York, NY

02/2025 – 05/2025

- Led all coding, data cleaning, and empirical strategy; estimated survey-weighted DiD and event-study models on CPS ASEC microdata; combined SUR models to test joint significance across coverage, financial protection, and health outcomes.

ADDITIONAL INFORMATION

Technical Skills

Proficient

R (Data Wrangling, ML, API Interaction, Imputation, Visualization, Statistical & Econometric Modeling); **Stata** (Data Wrangling, Imputation, Visualization, Statistical & Econometric Modeling)

Working Knowledge

Python (Data Wrangling, Web-scraping); **SQL** (Data Querying); **Tableau** (Visualization); **Qualtrics** (Survey Design); **LaTeX** (Formatting)

Public Datasets

NLSY, MEPS, NSFG, IPUMS (CPS, ATUS)

Nationality

Taiwan

Languages

English (Full Professional), Mandarin Chinese (Native)

Fun Facts

Former swimmer with 10+ years of competitive experience; National collegiate record holder and high-school athletic scholarship recipient in Taiwan. Avid Real Madrid fan and sports analytics enthusiast.

REFERENCES

Prof. Wen-Jui Han, Professor

NYU Silver School of Social Work

Prof. Mustafa Hussein, Assistant Professor

CUNY Graduate School of Public Health

Dr. Jui-Chung Allen Li, CEO

Quanthon Corporation

References available upon request.